

BLOOD DONATION

The Application Of Blockchain For Blood Donation



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AGENDA

I. CONTEXT

IV. APPLICATION - USE CASE

II. PROBLEM

V. WRAP-UP

III. BLOCKCHAIN - ANALYSIS

VI. REFERENCES



I. CONTEXT

Global (1/2)

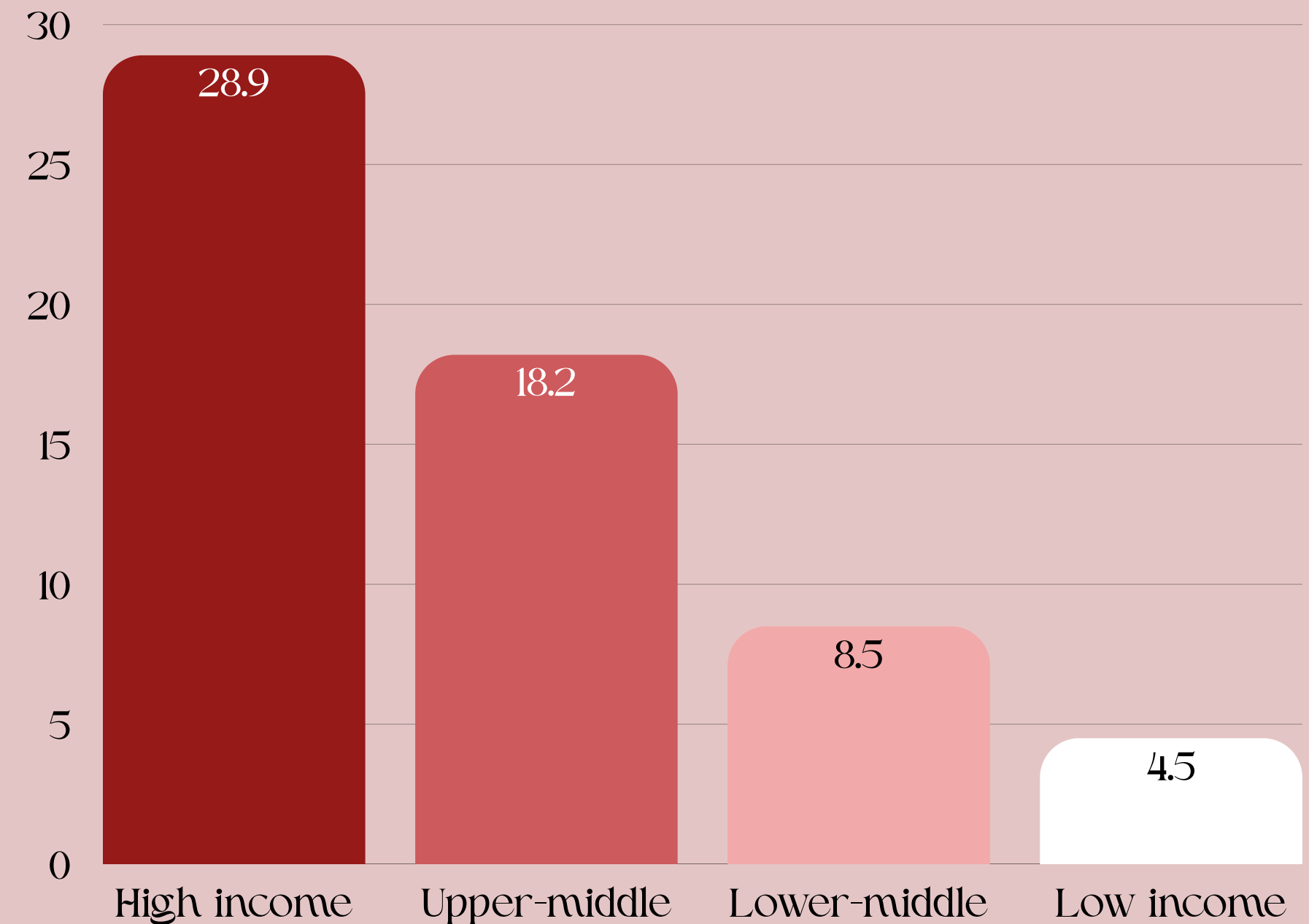
120+ MILLION

blood donations are collected every
year



More blood donations do not always
mean equal access to blood.

Median Blood Donation Rate per 1,000
Population



Source: WHO, 2026 (data: 2023)

I. CONTEXT

Global (2/2)

POSTPARTUM HEMORRAGE

around **45,000 women** die each year from severe bleeding after childbirth

SEVERE MALARIA IN CHILDREN

causes **over 600,000 deaths** each year, mainly among young children in Africa

TRAUMA AND ROAD ACCIDENTS

around **1.16 million people** die from road crashes each year



The real challenge is not only collecting blood, but making sure it is available when patients need it most.



Source: WHO, 2025. WHO – Postpartum haemorrhage
WHO, current page using 2024 estimates. WHO – Malaria
WHO, 2026. WHO – Road traffic injuries

I. CONTEXT

Vietnam (1/2)

1.75+ MILLION

units of blood collected annually

BUT

- Sudden disruptions in collection
- Fluctuations in demand
- Challenges in distribution



**Large-scale donation
≠
Reliable blood supply**

Source: Tôn Vinh Người Hiến Máu Tình Nguyên Tiêu Biểu Toàn Quốc
Năm 2025 – HỘI CHỮ THẬP ĐỎ VIỆT NAM, n.d.

I. CONTEXT

Vietnam (2/2)

A Critical Blood Shortage

COVID-19, 2021

Remain

13

platelet units

2,198

blood units

High Daily Demand
----> Limited Stock

Demand

≈ 200-300

units daily
demand

II. PROBLEM



GLOBAL: Fragmented Blood Information

- Separate information systems
- Limited interoperability
- Inconsistent data standards

Evidence: *The study examined **19 sites across 15 EU countries** (PETRA, 2026).*



VIETNAM: Fragmented Blood Data

- Separate systems across provinces
- Difficult data synchronization & retrieval
- Limited integration across healthcare organizations

Evidence: *NIHBT – **~4.3M donor records** in the national core system (NIHBT, 2026).*



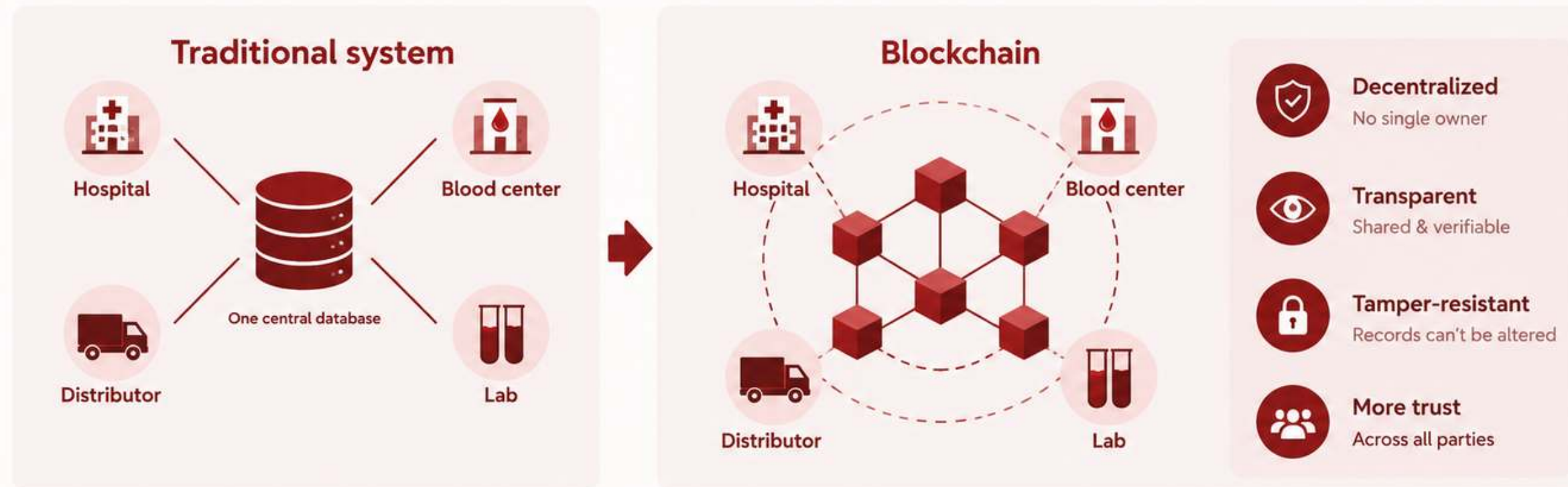
COMMON PROBLEM:

Fragmented and Disconnected Blood Information
→ Difficult to **SHARE, VERIFY & TRACE** blood information



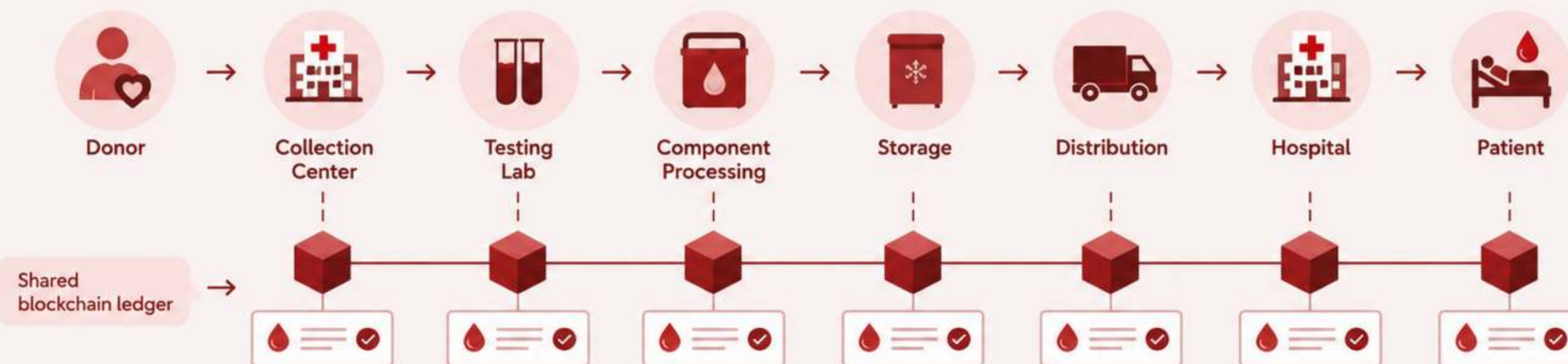
III. ANALYSIS (1/3)

Blockchain



In Blood Management

Blockchain connects the whole blood journey



01

DECENTRALIZED & SHARED LEDGER

02

SHARE & VERIFY INFORMATION

03

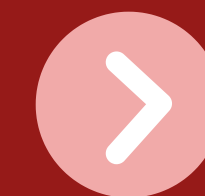
TAMPER-RESISTANT & TRANSPARENT

04

TRACE BLOOD UNITS ACROSS ORGANIZATIONS

05

CONNECTS HOSPITALS, BLOOD CENTERS & LABS



Blockchain connects fragmented blood information through a shared, verified, and traceable system.

III. ANALYSIS (2/3)

WHAT BLOCKCHAIN CAN SOLVE ?

01

SHARE

Connect fragmented information

Shared Ledger:
Connects authorized hospitals & blood centers.

Synchronized Access:
Shares donor records & blood inventory securely.

02

VERIFY

Protect data integrity

Cryptographic Verification:
Verifies the source of clinical records.

Data Integrity:
Makes unauthorized changes detectable.

03

TRACE

Track every blood unit

Unique Digital ID:
Links each blood unit to its digital record.

Auditable Chain:
Records the journey from donation → testing → storage → transfusion.

WHAT BLOCKCHAIN CANNOT SOLVE ?



1 Garbage In, Garbage Out

Wrong input → immutable wrong records
e.g., blood type, barcode, test results

2 Physical-Digital Disconnect

Cannot monitor temperature, location, or storage conditions

3 System Interoperability

Different HIS/LIS systems may not communicate seamlessly

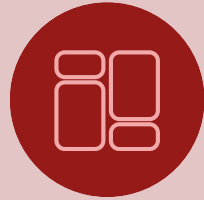
4 Privacy & Regulatory Conflicts

Immutability may conflict with data deletion & privacy requirements



III. ANALYSIS (3/3)

IV. Applications



Overview

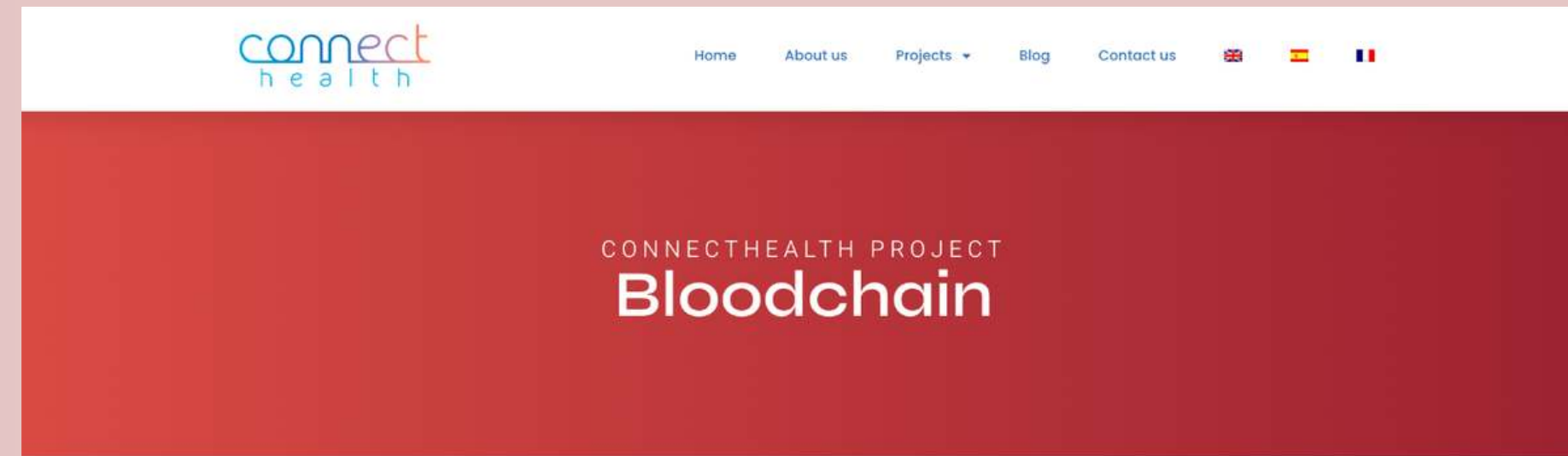
Initiated by ConnectHealth

Directly links donors, blood banks, and hospitals to optimize emergency matching and dispatch.



Blockchain

End-to-End Traceability: Serves as an immutable infrastructure tracking biological products across collection, processing, and clinical use.



Global Case (1/4)

Bloodchain

Traceability of biological products on blockchain.

Registration and identification of the products of biological origin (obtained from donors or other biological sources) and of different elements issued from the production/obtention and use of these (entities, human or animals) in order to facilitate the access and monitoring of the data in the entire chain of collection, production, supply and utilization of the products of biological origin for medicinal purposes in humans or other living beings, in a universal manner, from the origin (human or of another living being) until the final use of the product in humans or in other living beings, to certify anywhere, and in an universal manner, the origin, the destination or recipient, as well as the intermediaries and the quality control of the products.

1

**Bloodchain of
ConnectHealth**

IV. Applications



Value

Secure Donor Profiles

Stores verified donor screening records and blood typing on a tamper-resistant distributed ledger.

Data Privacy & Control

Empowers donors with ownership of their personal health data, permitting access only to certified medical facilities during donation.

Verified Eligibility

Eliminates repetitive pre-screening procedures by providing hospitals with trusted, immutable donor verification records.



Global Case (2/4)

1

Bloodchain of ConnectHealth



Bloodchain makes blood donation faster, safer, and more trustworthy.

2

BDSC Framework

(Blood Donation Supply Chain Framework)



Overview

A decentralized health logistics blueprint developed by Canadian researchers to modernize blood management using Ethereum.



Blockchain

Role-Based Smart Contracts

Governs access control across donors, blood banks, and hospitals.

Hybrid Storage

Logs unique tracking IDs on Ethereum while storing heavy clinical test files off-chain via IPFS.

IoT & AI Integration

Employs real-time cold-chain temperature tracking and predictive demand forecasting to minimize loss.





2

BDSC Framework

(Blood Donation Supply Chain Framework)



Value

Traceability

Prevents counterfeit records and cross-contamination in transfusions.

Reduced Wastage

Leverages real-time data to prevent stock expiration.

Donor Privacy

Protects personal identities through cryptographic hashing.



BDSC combines blockchain, IoT, and AI to make blood supply chains more efficient and reduce waste.



Vietnam Case: Giot Mau Vang (GMV)

Founders: Ho Chi Minh City Red Cross & Ho Chi Minh City Computer Association (HCA) launched in 2021

Project:

Digital Transformation to Improve the Management of Voluntary Blood Donation and Ensure Blood Transfusion Safety in Ho Chi Minh City

AS OF AUG, 2025

605,000+
registrations

100,000+/month
app visits

71.7%
repeat donor

Source: Nhân Dân News (2025)

Current Capabilities (Centralized Value)

- **Core Infrastructure:** Operates on a stable centralized database platform.
- **Tech Integration:** Leverages eKYC (citizen identity cards), AI algorithms, and a QR code scanning system to digitalize the donation workflow.



(2/4)

Phase 2 Vision (Planned Blockchain Objectives)

- **Unit Identification:** Utilizing data analytics and blockchain to assign a unique, immutable identifier to each blood bag.
- **End-to-End Integration:** Establishing immutable data connectivity across the entire workflow.

(Mobilization → Donation → Blood Bank → Hospitals)



Current Limitations

Challenge	Why it matters
Data Synchronization	VNeID and GMV still face integration & synchronization issues
Cost & Regulation	Blockchain requires operational resources and must address healthcare regulations
Privacy	Decree 13/2023/ND-CP grants citizens the "right to data erasure," conflicting with blockchain's immutable nature. <i>Source: Vietnam Government</i>



Managerial Recommendations

01

Define the strategic case - Should GMV pursue blockchain – and why?

- Strategic Value
- Ecosystem Readiness
- Value vs. Cost
- Governance

02

Design the ecosystem - What should the future blood-data ecosystem look like?

- Data & Interoperability
- Trust & Traceability
- Privacy & Governance

03

PILOT & SCALE - Can it create measurable value in practice?

Pilot → Measure → Scale

V. WRAP UP (1/2)

1. Key findings:

Blood Information Fragmentation

Fragmented blood information makes **sharing**, **verification**, and **traceability** difficult, even with over **120 million** blood donations collected globally each year.

Blockchain's Potential

Blockchain can strengthen data integrity, information sharing, and blood traceability through a shared ledger, verified records, and unique blood unit IDs.

Real-World Implementation

Existing applications show potential, but adoption depends on interoperability, infrastructure, and privacy compliance.

Next 

V. WRAP UP (2/2)

1. Key findings:

Vietnam's Readiness

Vietnam has an existing digital foundation, with over **1.75 million** blood units collected annually and **4.3 million** donor records, but data integration and standardization are still needed.

2. Final takeaway:

Blockchain is not a standalone solution. It should strengthen existing blood-management systems, supported by reliable data, connected infrastructure, and proper privacy protection.

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**THANK YOU FOR
YOUR ATTENTION!**